

AI ANALYTICS & TREND ANALYSIS DOCUMENTATION

1. Objective:

To use artificial intelligence as an analytical assistant to evaluate trends, performance patterns, and business metrics derived from the Power BI dashboard and Python statistical analysis, turning data into clear, structured business insights.

2. AI Prompt & Workflow Used:

After completing the Power BI dashboard and statistical analysis, the following prompt structure was provided to the AI to perform the analysis: "Based on our finalized Olist e-commerce project metrics (Total Revenue: R\$ 13.59M, Total Orders: 98,666, Repeat Customer Rate: 3.12%, On-Time Delivery Rate: 91.89%, and Python Statistical Price-Freight Correlation: $r = 0.4142$), analyze the trends and provide structured business insights and recommendations."

3. AI-Generated Trend Insights * Sales & Revenue Patterns:

The AI identified that marketplace revenue is heavily concentrated within top-performing product categories, showing that promotional budgets should focus on leading items. * Customer Retention Analysis: The AI highlighted the 3.12% repeat customer rate as proof that Olist operates primarily as a single-transaction marketplace, emphasizing the need for post-purchase engagement. * Operational & Statistical Performance: The AI analyzed the 12.56-day average delivery time, 7,827 late orders, and the moderate price-freight correlation ($r = 0.4142$) to highlight transit patterns and carrier optimization needs.